3.4.6 Wrap-Up: Ticket Out the Door

1. A line passes through the points $(-2.7, 0.78)$ and $(21, 15)$. Write the equation of the line in standard form.

Unit 3, Lesson 5: Scatter Plots and Linear Relationships

Benchmark

MA.912.DP.2.4 Fit a linear function to bivariate numerical data that suggests a linear association and interpret the slope and y-intercept of the model. Use the model to solve real-world problems in terms of the context of the data.

Learning Targets

- ☐ I can informally fit a straight line to model the relationship between two variables in a scatter plot.
- ☐ I can interpret the slope and y-intercept of a linear function fit to bivariate numerical data in context.
- ☐ I can determine an appropriate domain and range for bivariate numerical data based on the context.

3.5.1 Warm-Up: Bell Work

1. Complete the table by calculating the value that completes each statement given the context and identifying the slope of the line that represents the context.

Scenario	Unit Rate Statement	Slope of the Line Representing the Context
Five granola bars cost $\$20$.	A granola bar costs $_____.	
A car travels 100 miles at a constant speed for 2.5 hours.	The car is traveling _____ miles per hour.	
Tyler can do 50 sit-ups in 4 minutes.	On average, Tyler does _____ sit-ups per minute.	
3 ounces of yeast flakes cost $\$4.29$.	One ounce of yeast flakes costs $_____.	

3.5.2 Refresher: Scatter Plots and Their Data

Linear equations can also be used to model the relationship between ***bivariate data*** that shows a linear relationship.

Bivariate data is data that measures two characteristics of a population.

Bivariate numerical data can be represented using a ***scatter plot***.

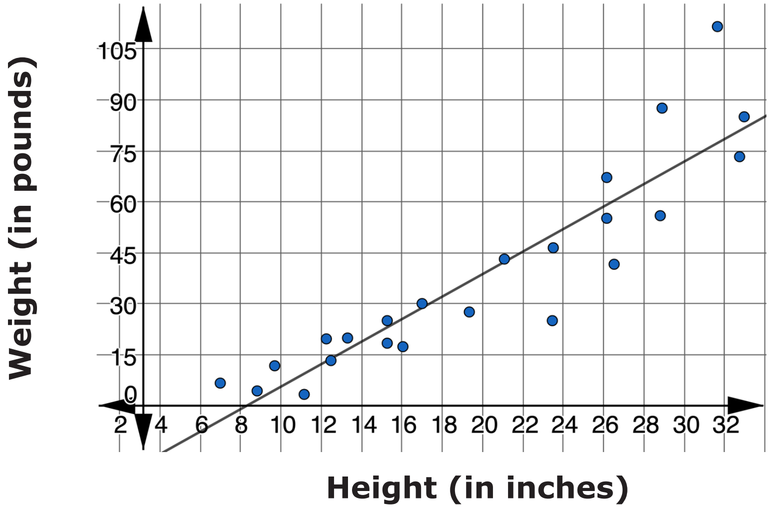
A ***scatter plot*** is a graph in the coordinate plane representing a set of bivariate numerical data that is used to observe the relationship between two variables.

Each point on a scatter plot represents a specific pair of values from the raw data. The x- and y-values are used to plot each data point in the coordinate plane. When all of the points in a data set are represented, a ***line of fit*** is often used to model the relationship between the two characteristics in the data set.

A ***line of fit*** is a line drawn on a scatter plot to estimate the relationship between two sets of data. A line of fit is also known as a trend line.

While different lines of fit can be drawn to model the relationship, the goal is to draw the line of **best** fit to most closely model the relationship and use it to make predictions.

1. The relationship between the height and weight of 23 dogs is shown.



Complete the statements using the terms in the bank. Some terms will not be used. Do not use a term more than once.

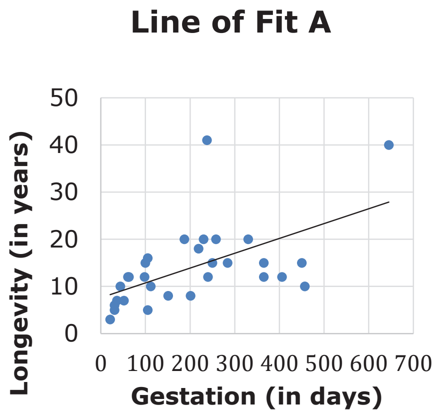
bivariate	decreases	increases
line of best fit	negative	positive
	scatter plot	

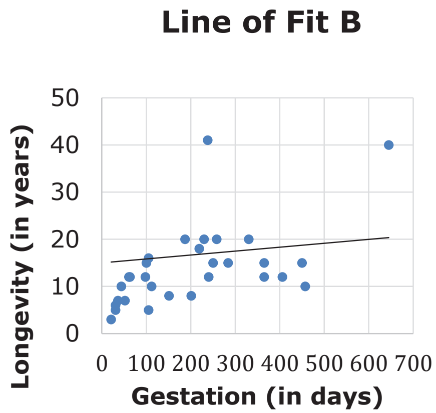
The ______________________________ shown above represents the relationship between ________________ data. The two variables being compared are the height and weight of 23 dogs. The trend line, or ______________________________, estimates the relationship between these two variables. For this relationship, as dog height increases, the weight of the dog _______________________, so the trend line has a _______________________ slope.

3.5.3 Collaboration: Scatter Plots and Patterns of Association

Work with your partner to complete the following.

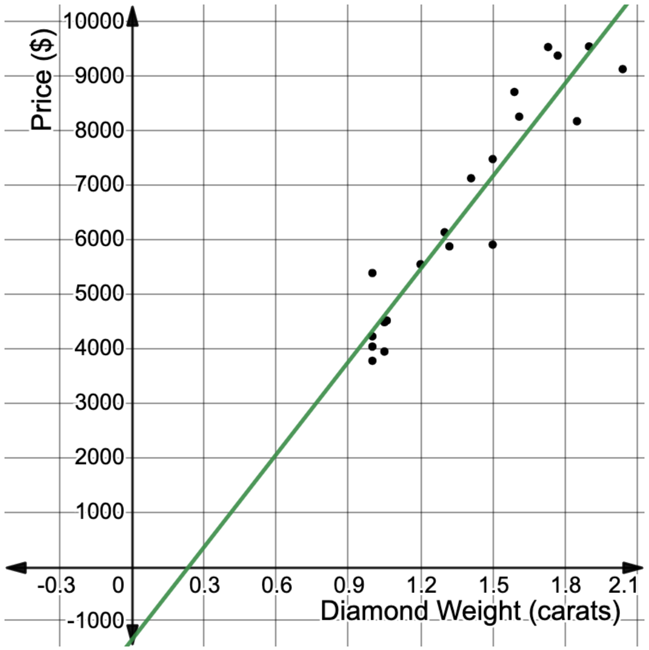
1. The two scatter plots show the same set of data summarizing the average gestation period, in days, and average longevity, in years, for a sample of 30 animals as reported in *The World Almanac and Book of Facts* (McGeveran, 2006). A gestation period is the amount of time for which an animal is pregnant. Longevity refers to how long an animal lives. Each scatter plot shows a different line to model the relationship between gestation period and longevity.





Discuss with your partner which scatter plot shows a line better fit to model the relationship. Summarize your discussion.

2. The scatter plot shows the relationship between the weight, in carats, and the price, in dollars, of 20 different diamonds.



The equation $y = 5{,}673x - 1{,}354$ models the relationship between a diamond's weight and its price based on the data shown. This line of fit can be used to predict the price of a diamond from its weight.

a. Define the variables in this context.

- x represents ______________________________
- y represents ______________________________

b. Use the model to predict the price of a diamond that weighs 1.5 carats.

c. Two of the diamonds from the original data set weighed 1.5 carats. The actual price of the two diamonds used to construct the scatter plot are shown in the table.

Weight (carats)	Actual Price ($)
1.5	7,474
1.5	5,904

Discuss with your partner how the actual prices of these two diamonds compare to the predicted price of a diamond of that weight. Summarize your discussion.

The line fit to the data in a scatter plot is used to model the ***association*** between the variables. It can be used to make estimates or predictions, but the actual values may fall above or below the line.

ABC

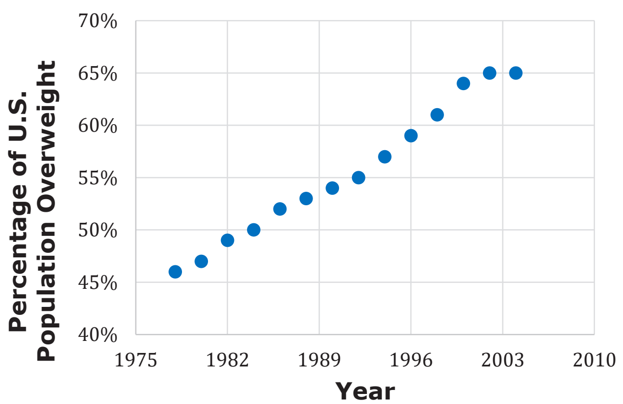
An ***association*** is a way to describe the form, direction, or strength of the relationship between the two variables in a bivariate data set. For numerical data, descriptions include linear or nonlinear; positive or negative; strong or weak.

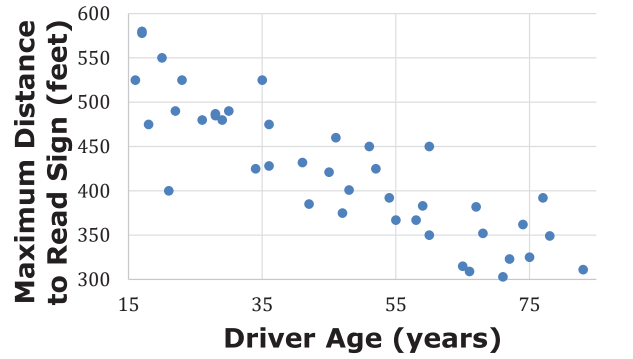
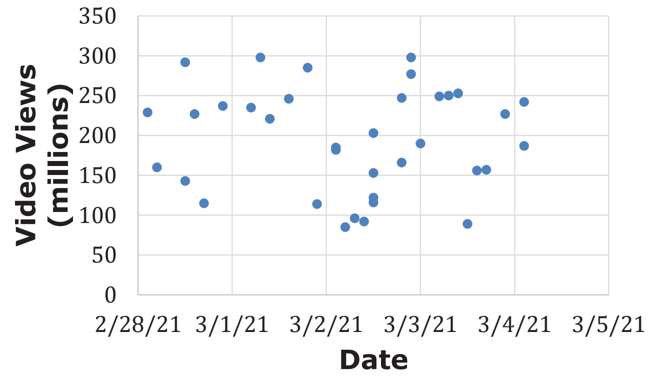
d. Complete the statements to describe the patterns of association for the scatter plot.

The relationship between a diamond's weight and price is

[○ linear. / ○ non-linear.] As a diamond's weight [○ increases, / ○ decreases,] the price [○ increases, / ○ decreases,] so the trend is [○ positive. / ○ negative.] The data points closely follow the line of fit, so the pattern of association is [○ weak. / ○ strong.]

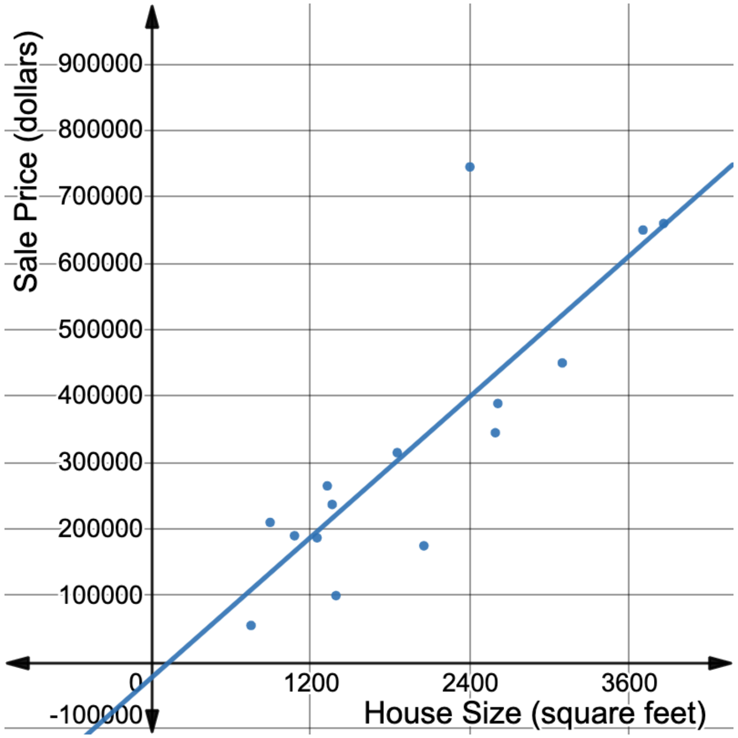
3. Complete the table to describe the pattern of association for each scatter plot.

Scatter Plot A	
Pattern of Association	☐ linear ☐ non-linear ☐ positive ☐ negative ☐ neither ☐ strong ☐ moderate ☐ weak

Scatter Plot B	
Pattern of Association	☐ linear ☐ non-linear ☐ positive ☐ negative ☐ neither ☐ strong ☐ moderate ☐ weak
Scatter Plot C	
Pattern of Association	☐ linear ☐ non-linear ☐ positive ☐ negative ☐ neither ☐ strong ☐ moderate ☐ weak

3.5.4 Guided Instruction: Interpreting the Slope and the y-intercept of the Line of Fit

1. The scatter plot shows the relationship between the house size, in square feet (ft^2), and the sale price, in dollars, of 16 houses sold in Orlando, FL in 2020.



The equation $y = 176.1x - 23{,}791$ can be used to model the relationship between size of a house, x, and its sale price, y, in Orlando, FL in 2020.

a. Determine the slope and y-intercept of the model.

Slope	
y-intercept	

b. Discuss with your partner what you think the slope means in this context.

c. Complete the statement.

In this context, the slope means the ○ house size ○ sale price ○ increases ○ decreases ○ $176.10 per square foot ○ 176.1 square feet per dollar for homes in Orlando, FL in 2020.

d. Trevor and Nadia both considered the meaning of the y-intercept in this context. Each student's thinking is shown in the table.

Trevor's Thinking	Nadia's Thinking
"The y-intercept means the sale price of a house with 0 square feet would have been $-\$23{,}721$ in Orlando, FL in 2020."	*"The meaning of the y-intercept does not make sense in this context. Houses don't have 0 square feet and they don't sell for negative amounts of money."*

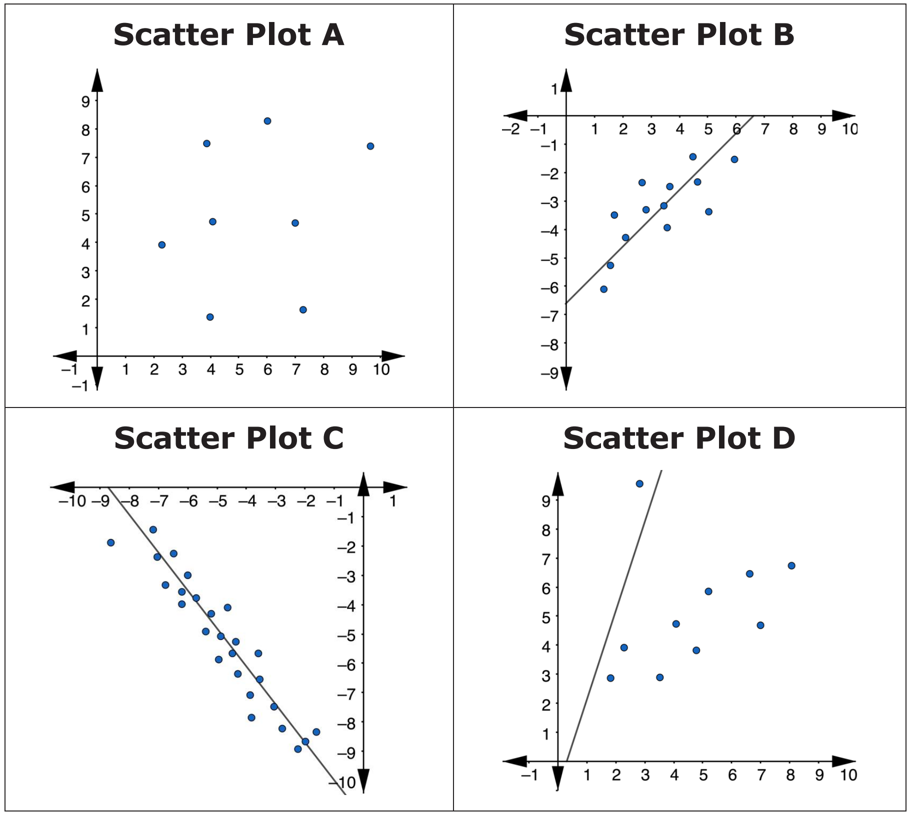
Discuss Trevor's and Nadia's thinking with your partner.

e. Determine an appropriate domain and range for the context, expressed as inequalities.

Domain	
Range	

3.5.5 Wrap-Up: Cool Down

Four scatter plots are shown.



1. Complete each statement to explain why one of the scatter plots doesn't belong with the others. Use a different scatter plot to complete each statement.

   a. Scatter plot ___ doesn't belong because ...

   b. Scatter plot ___ doesn't belong because ...

Unit 3, Lesson 6: Fitting Linear Functions to Data

Benchmark

MA.912.DP.2.4 Fit a linear function to bivariate numerical data that suggests a linear association and interpret the slope and y-intercept of the model. Use the model to solve real-world problems in terms of the context of the data.

Learning Targets

- ☐ I can formally fit a linear function to bivariate numerical data using technology.
- ☐ I can relate the correlation coefficient to the slope and fit of a linear relationship between two variables.